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Evaluating feedback mechanisms in the automatic creation of assembly instructions using retrieval-augmented generation

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Abstract

In recent years, Retrieval Augmented Generation (RAG) has emerged as a promising technology for designing application-specific chatbots. RAG combines a Large Language Model (LLM) with application-specific information provided via a database. This opens up the possibility for companies to at least partially automate existing business processes using RAG. One use case relates to work preparation at a mechanical engineering company. Due to the high complexity of the products and the wide variety of variants, creating assembly instructions for individual assemblies and modules involves a considerable amount of administrative work. The use of RAG can help to significantly improve the efficiency of this process. However, a key challenge is to ensure high quality of automatically generated assembly instructions. While existing research has focused on feedback mechanisms for improving LLMs, the effectiveness of different prompt-based feedback types on RAG-generated technical documentation quality has not yet been sufficiently investigated. This article therefore explores how quality of assembly instructions can be improved by integrating feedback mechanisms into a RAG system. For this purpose, two different feedback mechanisms were used in a laboratory experiment. On the one hand, an exclusively quantitative feedback mechanism was used. On the other hand, quantitative and qualitative feedback were combined. The results demonstrate that feedback mechanisms contribute to improving quality of information in assembly instructions. In addition, it was found that exclusively quantitative feedback leads to similarly good results as more complex combined feedback.

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Keywords: Retrieval-Augmented Generation; Feedback Mechanisms; Large Language Model; Assembly Instruction

1. Introduction

In recent years, significant progress has been made in the field of artificial intelligence (AI) with the help of large language models (LLMs) [1]. The basic functions and applications of an LLM are diverse. Key tasks include text generation and language comprehension [2]. Despite the high performance of existing LLMs, they are limited, especially when it comes to solving knowledge-intensive tasks. A central issue in this context is the hallucination problem, which refers to the tendency of LLMs to generate linguistically coherent but factually incorrect responses [3]. Among other things, hallucinations are generated when user queries go beyond the training data or require current information [3]. A promising solution is retrieval-augmented generation (RAG). RAG combines an LLM with application-specific information provided via a database [3]. Based on the structure of the first generation according to Lewis et al. [4], a so-called naive RAG comprises the steps of indexing, retrieval, and generation. Raw data is cleaned, segmented into chunks, vectorized using an embedding model, and stored in a vector database to enable efficient similarity comparison. The user query is also vectorized, compared with the segmented chunks, merged with the retrieved chunks to form a prompt, and then passed to the LLM. In generating the response, either only the

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aforementioned information is used, or the information base underlying the LLM is additionally used [3].

RAG has become a widely used technology for developing business- and domain-specific chatbots [3]. Against this background, new application possibilities are also opening up for manufacturing companies. These companies are increasingly confronted with the challenges of a growing number of product variants, shorter product life cycles, and the associated complexity [5]. This complexity leads to an increase in the amount of information that needs to be processed in a company. In terms of work preparation, this development means that the administrative expense required to provide workers with assembly instructions that meet the requirements for individual products is increasing. In this context, RAG systems offer the possibility of automating the process of generating assembly instructions. The quality of the generated assembly instructions is crucial for the acceptance of a RAG system. Incorrect instructions require that a work preparation employee has to revise them, resulting in the loss of at least some of the productivity gains achieved through the use of RAG.

In order to improve the quality of information output from LLM-based information systems, various approaches to integrating feedback are described in the literature. An overview can be found, for example, in Nathani et al. [10]. However, there are few empirical findings on prompt-based feedback in domain-specific applications. In particular, there is a lack of comparison between different prompt-based feedback mechanisms. Against this background, this article examines how feedback mechanisms can be used to gradually improve the quality of assembly instructions. The experimental design is based on the following central hypotheses:

- If different assembly instructions are generated one after the other using RAG, direct feedback on the information quality of each assembly instruction tends to increase the information quality of the generated assembly instructions.
- Combined feedback that includes both a qualitative and quantitative assessment of information quality tends to result in higher information quality in the assembly instructions generated than feedback that is exclusively quantitative.

2. Method

2.1 Experimental design

A naive RAG system is developed using the Microsoft Azure environment. The experimental design is shown in Fig. 1. ChatGPT-4o is used as the LLM. The prompt chaining method is used to generate assembly instructions. Two PDF files stored and vectorized in the data storage are used as the data basis for retrieval for each assembly. The first document contains an assembly BOM and information on the assembly sequence. In the second document, the tools required in the case of assembly are assigned to the components (tools-to-component relations). In this experiment, user feedback on the quality of the information output serves as the independent variable. A distinction is made between two types of feedback. In one case, the quality of the information output is evaluated purely quantitatively, and this evaluation is made available to the RAG system. In the other case, the quantitative feedback is supplemented by qualitative feedback. In this case, we also refer to combined feedback.

The first objective of this study is to determine whether feedback has a positive effect on the quality of subsequent information output (dependent variable). Second, the question of whether the two feedback mechanisms – quantitative and combined – have different effects on the dependent variable will be investigated.

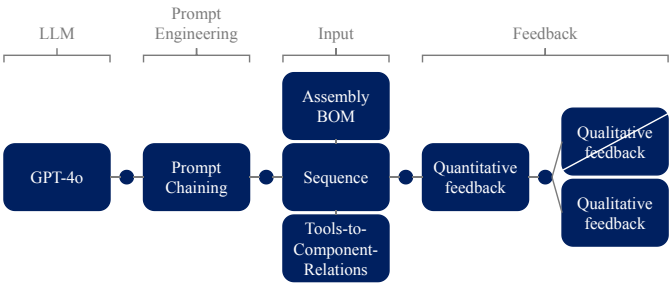


Fig. 1. Experimental design of this study.

Based on Herbort et al. [7], five different pneumatic assemblies, each consisting of 18 to 21 components are used for the experiment. Fig. 2 shows one of these assemblies with 18 components.

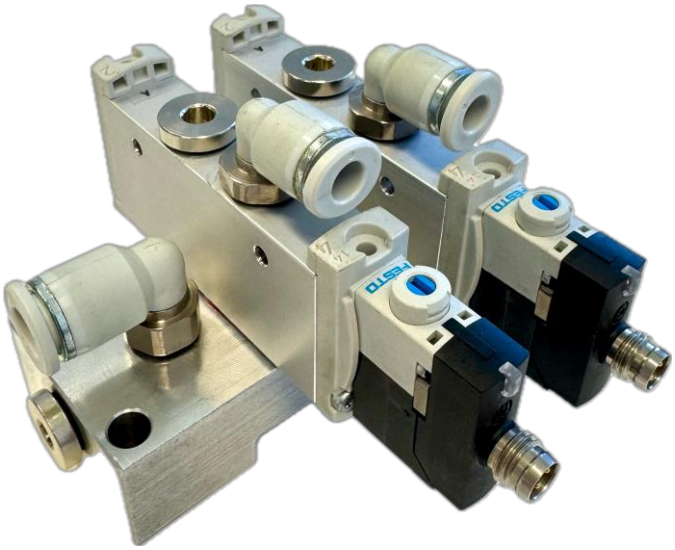


Fig. 2. Representation of one of the five pneumatic assemblies in the experimental procedure.

2.2 Criteria for evaluating information quality

In order to evaluate the quality of the information output, i.e., the generated assembly instructions, and to provide the RAG system with appropriate feedback, it is necessary to determine the requirements for assembly instructions and describe them using evaluation criteria. Assembly instructions have the purpose to select and present information on an assembly task in such a way that the sensory perception and processing of this information leads to effective (error avoidance) and efficient (short execution time) order processing, and at the same time, the support provided by the

instructions is perceived as highly useful by employees (high usability) [6]. Based on a description of the requirements for the design of assembly instructions [6], the following criteria for evaluating their quality can be derived: completeness, understandability, free of error, and appropriate sequence. Completeness means that all relevant information is included in the assembly instructions. Understandability means that the data can be clearly interpreted by the user. Free of error means that the assembly instructions do not contain any incorrect information. Appropriate sequence means that the assembly steps to be performed are listed in the correct order. The experiment assumes that assembly instructions contain descriptions of both the assembly process (assembly notes) and the tools (tool notes). Assembly notes are understood to be a list and description of the assembly steps to be carried out. Tool notes specify which tools are required to carry out the assembly task and how they are to be used. The quality of the information in the assembly instructions is therefore evaluated using the four criteria mentioned above, on the one hand for the assembly notes (see Table 1) and on the other hand for the tool notes (see Table 2). A four-point Likert scale is used in each case. The evaluation scheme is based on that of Herbort et al. [7].

Table 1. Criteria for evaluating the information quality of the assembly notes.

Criteria	1	2	3	4
Completeness	More than three required components are missing AND/OR the complete quantity specification is missing in the case of more than three components.	Two to three required components are missing AND/OR the complete quantity specification is missing in the case of two to three components.	At most one required component is missing AND/OR the complete quantity specification is missing for at most one component.	All required components including complete quantity information are specified.
Understandability	More than three implementation notes are ambiguous.	Two to three implementation notes are ambiguous.	At most one implementtation note is ambiguous.	All implementation notes are unambiguous and provide guidance for action.
Free of error	More than three component designations are incorrect.	Two to three component designations are incorrect.	At most one component designation is incorrect.	All component designations are correct.
Appropriate sequence	More than three steps are not listed in the correct order.	Two to three steps are not listed in the correct order.	One step is not in the correct order.	All steps are specified in the correct order.

Table 2. Criteria for evaluating the information quality of the tool notes.

Criteria	1	2	3	4
Completeness	More than three required tools are missing.	Two to three required tools are missing.	At most one required tool is missing.	All required tools are specified.
Understandability	More than three tool notes are ambiguous.	Two to three tool notes are ambiguous.	At most one tool note is ambiguous.	All tool notes are unambiguous.
Free of error	More than three tool designations are incorrect.	Two to three tool designations are incorrect.	At most one tool designation is incorrect.	All tool designations are correct.
Appropriate sequence	More than three tools are not listed in the correct order.	Two to three tools are not listed in the correct order.	At most one tool is not listed in the correct order.	All tools are listed in the correct order.

2.3 Feedback

For each experimental run, five different assembly instructions (for five different pneumatic assemblies) are generated via the RAG system. The quality of the first assembly instructions is evaluated using the schemes presented in Table 1 and Table 2, with one to four points being awarded per criterion according to the Likert scale. Since the assembly and tool notes are evaluated separately using four criteria each, the result is eight evaluations for one assembly instruction. Since a pretest showed that feedback from eight individual values is not effective, the mean value was calculated for each of the two values of a criterion (e.g., completeness) and rounded up. These four ratings were then bundled and made available to the LLM in natural language via user input.

In the case of combined feedback, qualitative feedback is transmitted to the RAG system in addition to the quantitative evaluation by providing standardized, text-based feedback (see Table 3) for each identified deficiency based on the evaluation criteria. This consists of the explicit mention of the deficiency and the justified classification of the deviation from the criterion. The aim is not only to point out a deficiency to the LLM but also to provide a justification for this classification.

Table 3. Examples of standardized, text-based feedback

Criteria	Example of text-based feedback
Free of Error	In step 3, “Assembly of subassembly BG3”, the tool designation “Open-end wrench (13 mm)” for the component “SLEEVE G 1/8 CUZN40PB2 NICKEL-PLATED (4-011-01-0075) / Quantity: 1” is incorrect. The correct tool designation would be “Open-end wrench (14 mm)”.
Appropriate sequence	Step 5: “Installing the large screw connector (angle)” is not listed in the correct order. Step 5 should be inserted after the step “Attaching the screw cap (large)”.

2.4 Procedure of the experiment

The prompt input represents the initial communication step with the RAG system. It includes information on the material number of the assembly and a request to generate the corresponding assembly instructions. The RAG system then generates step-by-step assembly instructions consisting of assembly and tool notes. The assembly instructions are then evaluated. Based on this evaluation, feedback is derived and transmitted to the RAG system as a response to the previous generation of the assembly instructions. The system confirms receipt of the feedback and the process is repeated with another assembly. This integration of feedback loops initiates an iterative process based on the principle of iterative refinement. The system generates five assembly instructions per experimental run, with feedback on the quality of the previous instruction being provided before the next instruction is generated.

In order to obtain representative results, four experimental runs are carried out for each of the two feedback mechanisms (quantitative and combined), resulting in a total of 20 assembly instructions (5 assemblies x 4 experimental runs) being generated and evaluated. To control for sequence effects, the trials were conducted twice in the original trial sequence (forward) and twice in the reverse trial sequence (backward).

2.5 Prompt Engineering

To generate assembly instructions, a structured prompt is formulated for each of the two feedback mechanisms and integrated into the Microsoft Azure AI Foundry system message. Prompt engineering deals with the development of prompts or instructions and has become an important success factor in the application of generative AI. It is a process of refinement that ensures that the output matches the user's expectations [8]. This study uses the so-called prompt chaining. This is a technique in which a chain of multiple tasks is created. By chaining together steps that build on each other, a higher probability of success can be achieved compared to solving complex prompts [9]. For both feedback variants, the prompt was divided into logically sequential sub-steps, ranging from the chat flow description to the step-by-step creation of the assembly instructions to the recording of feedback and the iterative refinement of the instructions based on this feedback.

3. Results

The quality of the assembly instructions was evaluated using the evaluation scheme for the elements “tool notes“ and “assembly notes“ using the four criteria and the four-point Likert scale. The test results are shown in a heat map. A rating of one point on the Likert scale is marked in red, while a rating of four points is highlighted in green. The results of the four test series on quantitative feedback are shown in Fig. 3, and the results on combined feedback are shown in Fig. 4. As can be seen from Fig. 3 and Fig. 4, the two elements of an assembly instruction – tool and assembly notes – were each evaluated using the four criteria of completeness, understandability, free of error, and appropriate sequence. The column “number of value '4'” (far right) shows how often the value “4“ (criterion fully met) was assigned to each generated assembly instruction. For each test series, the evaluation results for the first generated assembly instruction are listed in the top row, and the results for the last generated assembly instruction are listed in the bottom (fifth) row. The first two test series began with assembly no. 1 (forward), and the last two test series began with assembly no. 5 (backward). As shown in both figures, the number of ratings with the value “4“ tends to increase from the first to the fifth generated assembly (from the top row to the fifth row of a test series). This finding is illustrated by Fig. 5. This figure shows the mean value of the number of ratings with the value “4“ for each of the generated assembly instructions. A distinction is made between exclusively quantitative and combined feedback. In the case of both feedback mechanisms, it can be seen that the average number of ratings with a “4“ tends to increase from the first generated assembly instruction to the fifth generated assembly instruction, i.e., the quality of the information output tends to increase with each feedback loop. However, after four feedback loops, i.e., with the fifth assembly instruction, a fully requirement-compliant assembly instruction was not yet generated, since in this case all eight values for all four instructions would have had to be “4“ (instead of the average frequency of 8, only an average frequency of 5.25 and 6.75 was achieved).

Experiment						Key elements of assembly instructions								Number of value '4'
LLM	Input	Feedback mechanism	Experiment sequence	Series of experiments	Assembly	Tool notes	Assembly notes	Tool notes	Assembly notes	Tool notes	Assembly notes	Tool notes	Assembly notes	
						Completeness		Understandability		Free of error		Appropriate sequence		
ChatGPT-4o	Assembly BOM, Sequences and Tools-to-Component-Relations	Quantitative	Forward	1	1	4	4	4	2	3	3	3	4	4
					2	3	2	3	2	3	4	3	4	2
					3	4	4	4	2	4	4	4	4	7
					4	4	3	4	4	4	4	3	3	5
					5	2	4	4	4	4	4	4	4	7
				2	1	4	3	4	2	4	4	3	4	5
					2	3	3	4	1	4	4	3	4	4
					3	3	1	4	2	4	4	4	3	4
					4	4	3	4	1	4	4	4	3	5
					5	3	4	4	3	4	4	4	4	6
			Backward	3	5	2	4	4	3	4	3	3	4	4
					4	4	3	4	4	4	4	4	4	7
					3	2	4	4	4	4	4	2	4	6
					2	2	4	4	3	4	4	3	4	5
					1	3	4	4	4	4	4	4	4	7
				4	5	4	4	2	2	4	4	1	1	4
					4	4	4	4	3	4	4	4	4	7
					3	4	4	4	4	3	4	4	3	6
					2	2	4	4	4	4	4	4	4	7
					1	4	3	4	4	4	4	4	4	7

Fig. 3. Test results with integration of quantitative feedback.

Experiment						Key elements of assembly instructions								Number of value '4'
LLM	Input	Feedback mechanism	Experiment sequence	Series of experiments	Assembly	Tool notes	Assembly notes	Tool notes	Assembly notes	Tool notes	Assembly notes	Tool notes	Assembly notes	
ChatGPT-4o	Assembly BOM, Sequences and Tools-to-Component-Relations	Quantitative and qualitative	Forward	5	1	3	2	4	3	4	4	3	2	3
					2	3	2	4	3	4	4	1	1	3
					3	4	2	4	3	4	4	3	2	4
					4	4	4	4	4	3	4	3	1	5
					5	3	4	4	4	4	4	4	4	7
				6	1	4	1	4	3	3	4	4	4	5
					2	3	4	4	1	2	4	4	4	5
					3	4	4	4	3	4	4	4	4	7
					4	4	4	4	3	4	4	4	2	6
					5	3	2	4	4	2	4	4	4	5
			Backward	7	5	4	3	4	3	4	4	1	1	4
					4	4	4	4	3	4	4	4	2	6
					3	4	4	4	3	4	4	4	4	7
					2	3	2	4	2	3	4	3	4	3
					1	2	2	4	3	4	4	3	2	3
				8	5	3	4	4	3	4	4	1	4	5
					4	3	4	3	4	4	4	2	4	5
					3	4	4	4	4	4	4	3	4	7
					2	2	1	4	3	4	4	1	4	4
					1	4	4	3	4	4	4	2	4	6
						Completeness		Understandability		Free of error		Appropriate sequence		

Fig. 4. Test results with integration of combined feedback.

When comparing the two feedback mechanisms, it can be seen that without feedback influence (generation of the first assembly instruction), there is no difference in the average frequency of the "4" rating. However, the mean values for the subsequently generated assemblies differ. With quantitative feedback, the highest value (Ø 6.75), i.e., the best information quality, is achieved with the generation of the fifth assembly instruction, while combined feedback achieves the highest information quality (Ø 6.25) after two feedback loops, i.e., in the case of the generation of the third instruction.

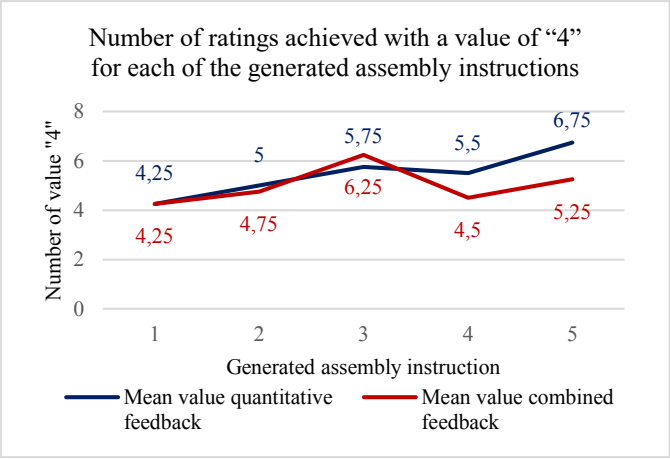


Fig. 5. Performance comparison of feedback mechanisms.

4. Discussion

The results show that feedback (independent variable) tends to have a positive effect on the quality of subsequent information output (dependent variable). This finding supports the validity of the first hypothesis (see section 1). However, the second hypothesis, according to which combined feedback is superior to exclusively quantitative feedback, must be rejected. As can be seen from Fig. 5, exclusively quantitative feedback actually leads to a slightly higher overall information quality than combined feedback. It is noticeable that with combined feedback – after the feedback on the third assembly instruction – the information quality deteriorates again (see Fig. 5). The reason for this deterioration can be attributed to a process known as over-refinement [10], i.e., too much feedback information may be confusing and, as a result, leads not to the expected improvement in the quality of the information output, but to the opposite, a deterioration.

The results for the individual evaluation criteria show that the feedback mechanisms did not affect the criterion of free of error, as the criterion was already largely fulfilled in the first assembly instruction generated, leaving little room for improvement. In contrast, the three other criteria proved to be more susceptible to content-related deficiencies. In particular, the criterion of correct appropriate sequence proved to be particularly prone to errors in the case of the combined feedback mechanism.

5. Limitations

One limitation arises from the fundamental limitations of naive RAG approaches, as already described by Gao et al. [3]. For example, a single retrieval is prone to low precision, which means that relevant content is overlooked and irrelevant text excerpts are included. In the generation phase, there is a risk of hallucinations and the output of irrelevant or distorted information.

The two feedback mechanisms, based on the quantitative and qualitative evaluation of the previous information output, also have limitations. Quantitative feedback is based on the rounded mean value for the elements "tool notes" and "assembly notes". The rounding up and the lack of differentiation between "tool notes" and "assembly notes" can lead to distorted results. The limitation of qualitative feedback is that the feedback can be very extensive in cases where the score is low, which could make it difficult to process this information.

Another point of criticism of the chosen methodology is the small number of only four test series carried out per feedback mechanism. This limits the significance of the improvements within the test results. Furthermore, it should be noted that the manual evaluation and entry of feedback came with a high expense, which limits the scalability of the evaluation method presented. In addition, due to the use of a Likert scale, ordinal scaled data is available, which can only be statistically analyzed to a limited extent.

6. Conclusion

This study contributes to the theoretical and empirical foundation of effective feedback procedures with regard to improving the quality of information output from RAG systems. Important findings of the study are, first, that feedback

mechanisms can, in principle, contribute to increasing the quality of information output. Second, the study shows that comprehensive feedback (quantitative and qualitative feedback) does not necessarily lead to higher quality output information than simple feedback in the form of ordinal-scale data (comparable to school grades). For example, the comprehensive, combined feedback initially resulted in significant improvements (up to the assembly instructions for assembly 3), but this level of quality could not be maintained thereafter. In particular, the criterion of the appropriate sequence of work steps proved to be particularly prone to errors in the combined feedback mechanism. The central finding of this study is that the quality of automatically generated assembly instructions using RAG can be improved through the targeted integration of feedback mechanisms. The improvement also depends on the type of feedback.

A promising approach for future research lies in the investigation of modular RAG architectures, which potentially enable further quality improvements. In addition, based on previous findings, the integration of multi-agent systems can also be a useful option. On the one hand, these offer the potential to support the further development of a specialized feedback agent. On the other hand, they open up the possibility of optimizing the automatic generation of assembly instructions through the division of labor in the processing of various subtasks. Such an extension would not only increase the flexibility and scalability of the overall system but also open up new potential for optimizing complex generation processes.

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